



GREENTECH

SMART GREENHOUSE SYSTEM

An IoT-enabled, AI-assisted greenhouse platform that helps Kenyan farmers monitor, control, and optimize growing conditions in real time.

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1. Background Information

Agriculture is a cornerstone of Kenya's economy, underpinning livelihoods, food security, and national growth. Yet farmers face mounting pressure from climate variability, inefficient resource use, rising input costs, and stagnating yields. Traditional farming methods lack the precision and adaptability needed to respond to these pressures, resulting in avoidable waste and inconsistent output.

Overuse of water & fertilizer

Manual application leads to significant resource waste.

Poor environmental control

Greenhouses lack the tools to regulate conditions precisely.

High labor dependency

Manual monitoring and intervention drive up operating costs.

Inconsistent yields

Unmanaged variability in conditions reduces productivity.

Advances in the Internet of Things (IoT), automation, and data analytics now make it possible to address these gaps directly. **GreenTech** is a fully automated, intelligent greenhouse system that lets farmers monitor, control, and optimize growing conditions in real time — improving productivity while using resources more sustainably.

2. Statement of the Problem

Kenyan farmers face several compounding challenges in greenhouse farming:

- **Resource inefficiency** — overuse of water, fertilizer, and energy drives up costs and harms both crops and the surrounding ecosystem.
- **Low productivity** — yields are limited by human error, unchecked pest infestations, and weak environmental control.
- **Rising input costs** — volatile prices for seeds, fertilizer, and skilled labor place farmers under financial strain.
- **Food insecurity** — climate unpredictability and population growth threaten long-term agricultural sustainability.

Together, these inefficiencies produce low productivity, high operating costs, and heightened vulnerability to climate change — underscoring the need for a smart, automated greenhouse system that combines real-time monitoring, intelligent control, and predictive analytics.

3. Research Objectives

General Objective

To design, develop, and deploy a smart automated greenhouse system that enhances agricultural productivity through precision farming technologies.

Specific Objectives

- Design a smart greenhouse system architecture integrating IoT, automation, and cloud-based technologies.
- Develop an IoT-based real-time environmental monitoring system.
- Design and implement automated control of irrigation, ventilation, and lighting.
- Develop a mobile and web-based application for remote monitoring and control.
- Integrate data analytics and AI algorithms for intelligent decision-making.
- Design and implement a smart pest-detection and targeted-spraying system.
- Pilot the GreenTech system under real farming conditions and evaluate its performance.
- Conduct a cost-benefit analysis to assess the system's economic viability.

4. Justification

GreenTech aligns strongly with national and global development priorities:

Kenya Vision 2030

Supports agricultural transformation, food security, digital innovation, and value-chain modernization.

African Union Agenda 2063

Promotes inclusive agricultural growth, technological advancement, and improved livelihoods.

BETA Model

Strengthens agricultural productivity, agribusiness innovation, and digital inclusion for farmers.

National priorities

Addresses ICT, health, environment, energy efficiency, and agribusiness innovation goals.

Sustainable Development Goals

SDG 2 · Zero Hunger

SDG 3 · Good Health & Well-being

SDG 8 · Decent Work & Economic Growth

SDG 9 · Industry, Innovation & Infrastructure

SDG 12 · Responsible Consumption & Production

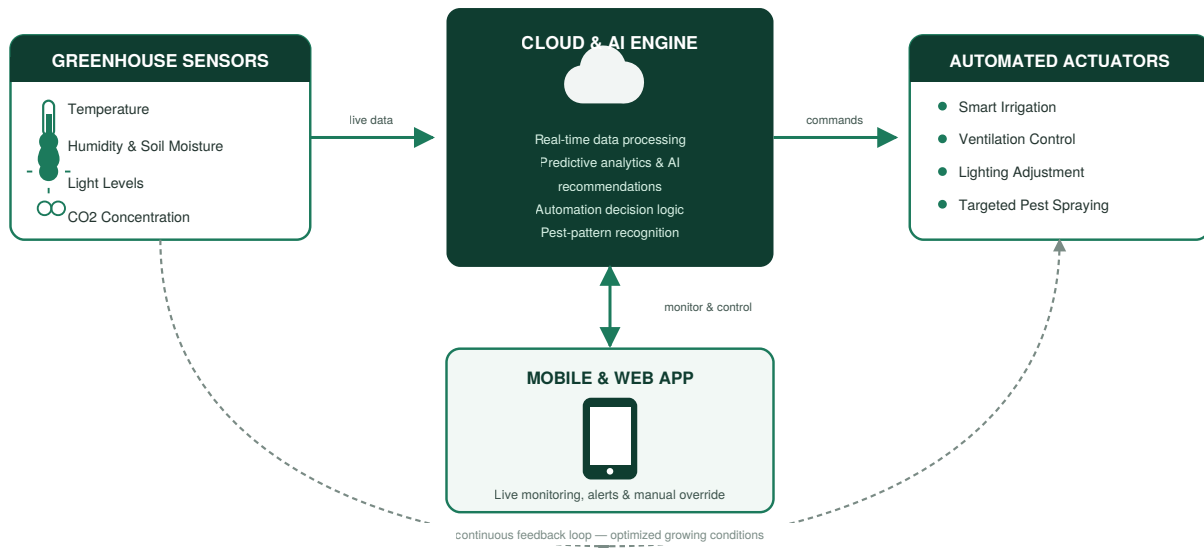
SDG 13 · Climate Action

5. Significance

- **Enhances agricultural productivity** — enables precise control of temperature, humidity, and soil moisture.
- **Promotes efficient resource use** — water, fertilizer, and energy are applied only when necessary, through sensor-based automation.
- **Reduces labor dependency and human error** — automating irrigation and ventilation lowers reliance on manual labor.
- **Supports climate-smart agriculture** — a controlled environment mitigates droughts, excess rainfall, and temperature swings.
- **Improves farmer income** — higher productivity and lower input costs directly boost profitability.
- **Enables data-driven decisions** — improves food availability at household and national levels.
- **Drives technological innovation** — promotes adoption of IoT, AI, and automation across the agricultural sector.

6. How GreenTech Works

GreenTech continuously senses greenhouse conditions, makes automated decisions in the cloud, and acts on the environment — while keeping the farmer informed and in control from a mobile app.



Core Functionalities

1. Real-Time Environmental Sensing

Sensors continuously track temperature, humidity, soil moisture, light, and CO₂ levels throughout the greenhouse.

2. Automated Climate & Resource Control

Irrigation, ventilation, and lighting are triggered automatically based on live sensor readings — no manual intervention needed.

3. Smart Pest Detection & Spraying

The system identifies pest activity early and applies targeted spraying, reducing crop loss and chemical overuse.

4. AI-Powered Decision Support

Predictive analytics recommend irrigation schedules, planting adjustments, and yield-optimization actions.

5. Remote Monitoring & Control

A mobile and web dashboard lets farmers view live conditions and override automated actions from anywhere.

6. Smart Alerts & Reporting

Farmers receive instant notifications on anomalies, alongside historical reports on yield, efficiency, and cost savings.

7. Financial Plan

7.1 Budget Summary

#	Budget Category	Description	Amount (KSh)
1	Pilot Development (Hardware & Greenhouse Setup)	Greenhouse structure, IoT devices, automation systems	520,000
2	Software & System Development	Mobile app, web dashboard, backend & cloud infrastructure	330,000
3	Deployment, Training & Marketing	Pilot rollout, logistics, training, awareness campaigns	450,000
TOTAL			1,270,000

7.2 Pilot Development (Hardware & Greenhouse Setup)

Item	Description	Cost (KSh)
Greenhouse Structure Materials	Frame, UV polythene, support structures	200,000
Irrigation System	Pipes, drip system, water pump, valves	90,000
Ventilation System	Fans, air circulation components	50,000
IoT Sensors	Temperature, humidity, soil moisture, light, CO ₂	70,000
Microcontrollers & Electronics	Raspberry Pi boards, wiring, power units	40,000
Installation & Assembly	Technical labor and setup	40,000
Miscellaneous Hardware	Connectors, fittings, backup components	30,000
Total		520,000

7.3 Software & System Development

Item	Description	Cost (KSh)
Mobile Application Development	Android app (monitoring & control)	120,000
Web Dashboard Development	Admin panel and analytics interface	80,000
Backend Development	APIs, database integration, system logic	60,000
Cloud Infrastructure Setup	Hosting, database, storage	40,000

Total	300,000
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Note: source figures list this category at 300,000 in the itemized breakdown and 330,000 in the summary total above; both are reproduced here as provided.

7.4 Deployment, Training & Marketing

Item	Description	Cost (KSh)
Pilot Installation	Setup in selected farms	150,000
Logistics & Transport	Equipment delivery and movement	80,000
Farmer Training Workshops	Training materials, facilitation	90,000
Marketing & Awareness Campaigns	Branding, outreach, campaigns	70,000
Operational Costs	Field support, maintenance	40,000
Community Engagement	Demonstrations, stakeholder meetings	20,000
Total		450,000

8. Go-To-Market Strategy

8.1 Greenhouse Product Pricing

Product	Description	Features Included	Price (KSh)
Standard Greenhouse	Basic, non-automated greenhouse	Manual irrigation, basic structure, no sensors	120,000
Smart Greenhouse	Fully automated & IoT-enabled	Sensors (temp, humidity, soil moisture, light), smart irrigation, ventilation, lighting, pest control, AI recommendations	300,000

The Standard Greenhouse is a one-time purchase requiring no subscription. The Smart Greenhouse is also a one-time hardware purchase — including all sensors and automation hardware — with its intelligent features unlocked through a monthly subscription.

8.2 Smart Greenhouse Subscription Plans

Plan	Features Enabled	Price (KSh / Month)
Basic	Remote monitoring, basic alerts (SMS/app), limited dashboard access	1,000
Standard	Full dashboard access, smart alerts, historical data & reports, basic automation optimization	2,500
Advanced	AI recommendations (irrigation, planting, yield), smart irrigation scheduling, predictive analytics, priority support	5,000
Premium	All Advanced features, pest-detection AI, advanced climate & lighting automation, dedicated support	10,000

8.3 Example Revenue Projection

Based on an initial base of 50 Smart Greenhouse buyers distributed across subscription tiers:

Plan	Farmers	Avg. Subscription (KSh)	Monthly Revenue (KSh)	Yearly Revenue (KSh)
Basic	15	1,000	15,000	180,000
Standard	15	2,500	37,000	444,000
Advanced	10	5,000	50,000	600,000
Premium	10	10,000	100,000	1,200,000

Total	161,250	1,935,000
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Figures reproduced as provided in the source projection.